

Faster R-CNN 리뷰

BOAZ 분석 24기 이권

Faster R-CNN.

Region Proposal Network. 기존 Fast-RCNN. 33.

ⓐ Selective Search 33. PRN → Pool 7x7.

과정 ① raw data. → ② Pre-trained CNN. → feature map 33.
 ③ feature map → PRN. → region proposals 11x11.
 ④ ②+③ 과정이후 생성된 feature map → Pool pooling. → fixed feature map.
 ⑤ ④로 인해 fixed feature map 일러 → classification & Bounding box regression 가능.

feature extraction by pre-trained VGG-16.
 사전 학습된 VGG-16 모델. $800 \times 600 \times 3$ → $50 \times 50 \times 4/2$.
 feature map.

Generate Anchors by Anchor generation layer.
 원본 이미지 크기 $(800 \times 600 \times 3)$ x sub-sampling ratio = $\boxed{\text{grid cell}}$.
 grid cell 17x17 9x9 anchor box 생성.

Class Scores and Bounding box regressor by PRN.
 PRN → VGG 16-33 layer feature map 일러 anchor의
 class score, bounding box regressor 출력.

Select anchors for training PRN by anchor target layer.
 Positive sample - 객체가 존재하는 영역 foreground, negative sample - 객체가 없는 영역 background.

select anchors for training Fast R-CNN by proposal

Target layer. $\rightarrow$ proposal layer에서 region proposal을

R-CNN과 같이 학습을 할 수 있는 sample (예)

$\rightarrow$ region proposal을 feature map에 ROI pooling을

max pooling by ROI pooling.

$\hookrightarrow$ fixed feature map을

Train F-R-CNN by multi-task loss.

Fast R-CNN은 공유된

feature vector를 classifier와 Bounding box regression

을. $\rightarrow (k+1) \times (1011) \times 4$

(feature vector를)

$\hookrightarrow$ Multi-task loss를

Fast R-CNN을

Loss Function.

$$L(p_i, (t, s)) = \frac{1}{N_{cls}} \sum_i L_{cls}(p_i, p_i^*) + \gamma \frac{1}{N_{reg}} \sum_i p_i L_{reg}$$

$i \rightarrow$ one data.

$p_i \rightarrow$ probability.

(t_i, s_i^*)

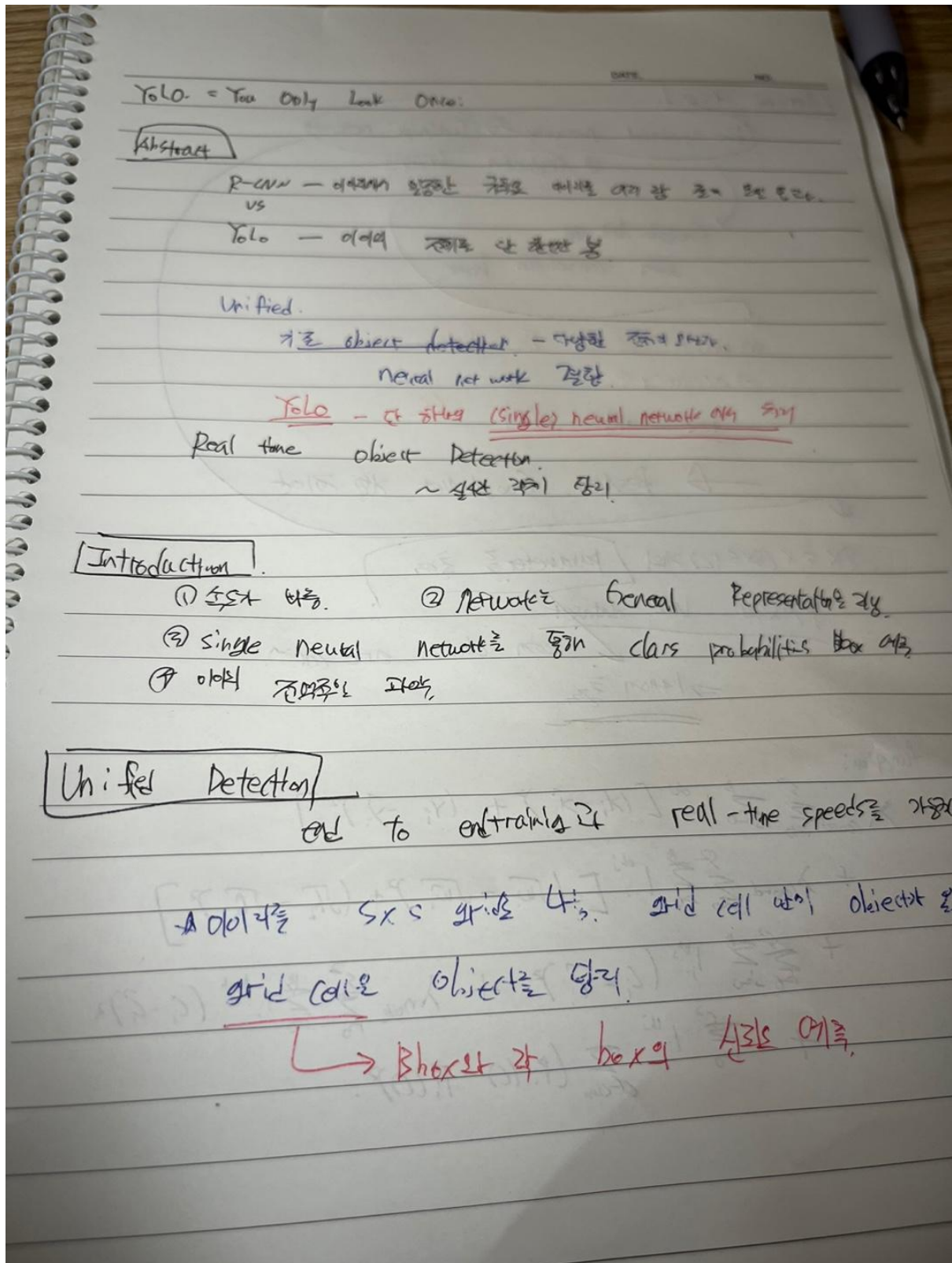
또한 Regression loss.

smooth loss

$$\text{Smooth}_i(x) = \begin{cases} 0.5x^2 & |x| < 1 \\ |x| - 0.5 & \text{else} \end{cases}$$

YOLO 리뷰

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Network Design

Pre-trained network + Training network
+ Reducer layer

GoogleNet is
2011 convolution layer

Network is so big feature maps are
→ accuracy ↑.

but it's slow

→ ResNet, GoogleNet etc not

5X5X (6x5 FC) not parameter is too big

PASCAL VOC dataset

2011 class. and cell

→ 141011

loss function:

$$\lambda_{coord} \sum_{i=0}^S \sum_{j=0}^B l_{ij}^{obj} [(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2]$$

$$+ \lambda_{coord} \sum_{i=0}^S \sum_{j=0}^B l_{ij}^{obj} [(\sqrt{x_i} - \sqrt{\hat{x}_i})^2 + (\sqrt{y_i} - \sqrt{\hat{y}_i})^2]$$

$$+ \sum_{i=0}^S \sum_{j=0}^B l_{ij}^{cls} (C_i - \hat{C}_i)^2 + \lambda_{mask} \sum_{i=0}^S \sum_{j=0}^B l_{ij}^{mask} (C_i - \hat{C}_i)^2$$

$$+ \sum_{i=0}^S l_i^{cls} \sum_{C \in class} (P_i(C) - \hat{P}_i(C))^2$$